

DV SERIES ■ LOW IMPEDANCE, 105°C TYPE

KEY FEATURES

- ALUMINUM ELECTROLYTIC CAPACITOR ■ SMD type
- Low impedance and long life



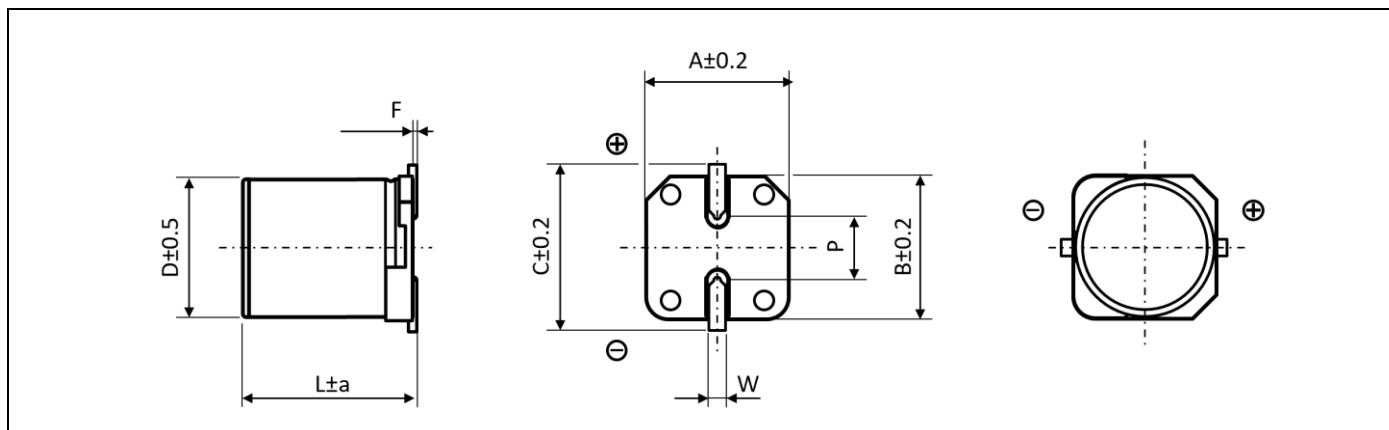
SPECIFICATIONS

Part Number		DV681M6R3F105ETRU		Unit	Conditions
Capacitance	C_R	680		μF	
Rated Voltage	V_R	6.3		V DC	
Surge Voltage	V_S	7.2		V DC	
Cap. Tolerance	ΔC	-20~20		%	120Hz ■ 20°C
Ripple Current	I_R	325		mA	120Hz ■ 105°C
Ripple Current	I_R	500		mA	100KHz ■ 105°C
Dissipation Factor	$\tan\delta$	24		%	120Hz ■ 20°C
Leakage Current	I_{LEAK}	43		μA	2 min ■ 20°C
Max. ESR	ESR	0.25		Ω	100KHz ■ 20°C
Size	D x L	8X10.5		mm	
Operating Temperature Range		-55~105		°C	
AEC-Q200 Qualified		NO			
Vibration Proof Version		NO			
Lifetime Test					
Endurance V_R applied	Test	2000		h	105°C
	$\Delta C/C_R$	$\leq$	± 30	%	of initial measured value
	$\tan\delta$	$\leq$	300	%	of initial specified value
	I_{LEAK}	$\leq$			the initial specified value
Shelf Life $V_R = 0$	Test	1000		h	105°C
	$\Delta C/C_R$	$\leq$	± 30	%	of initial measured value
	$\tan\delta$	$\leq$	300	%	of initial specified value
	I_{LEAK}	$\leq$			the initial specified value
Before measurement: Restore capacitor to 20°C, apply V_R for 30 min according JIS-C-5101-4					
Resistance to Soldering Heat	$\Delta C/C_R$	$\leq$	± 10	%	of initial measured value
	$\tan\delta$				Less than specified value
	I_{LEAK}				Less than specified value
The capacitors shall be kept on a hot plate maintained at 250°C for 30 seconds. After removing from the hot plate and restored at room temperature, they meet the characteristic requirements listed below					

PART NUMBER SYSTEM

Series	C_R (μF)	ΔC (%)	V_R (V)	D (mm)	L (mm)	Type	Lead Treatment	Special
1_2	3_4_5	6	7_8_9	10	11_12_13	14	15_16	17_18_19_20
DV	681	M	6R3	F	105	E	TR	U
	680	-20~20	6.3	8	10.5	With lead treatment	Process code	U=Package

DIMENSIONS ■ In mm



$\varnothing D$	L	a	A	B	C	F	W	P ± 0.2
8	10.5	max	8.3	8.3	9	0.3 max.	0.7~1.1	3.1

MULTIPLIER K_f for RIPPLE CURRENT vs. FREQUENCY

Frequency (Hz)	60	120	300	400	500	1k	3k	10k	50k	100k	300k
K_f	0.52	0.65	0.65	0.65	0.8	0.89	0.89	0.97	1	1	1

PRECAUTIONS, GUIDELINES AND PACKAGING INFORMATION

Unless otherwise agreed in individual specifications, all products are subject to our “General Precautions and Guidelines” as well as our “Packaging Information”. Please refer to the following links in the table.

General Precautions & Guidelines	Packaging Information	Vibration Test Profiles	3D Models	Reliability Tests

DISCLAIMER

All product related data (e.g. specification, statements and general information) are subject to change without any notice. It is necessary that the customer observes all product related technical / application information and handling instructions.

CapXon products are designed and manufactured according to severe quality and safety standards. Under no circumstance, CapXon warrants that any CapXon product is suitable for the purposes intended for your application, even CapXon knows the application. It is customer's duty and obligation to check and make sure that CapXon products are suitable for the purposes intended and select the correct and proper CapXon product. Customers are requested to perform a sufficient validation and reliability evaluation to assure needed safety level and reliability performance by suitable designs and to apply proper safeguards (e.g. redundancies, protective circuits).

Particular operating conditions (ambient temperature, ripple current, voltage, thermal resistance, etc.) as well as storage, production or assembly may affect the performance and the lifetime of the capacitor. Please consult CapXon for lifetime estimation, failure mode considerations or worst-case scenarios according to the product technology, product tolerances / deviations or change of the characteristics of the capacitor due to shipment, storage, handling, production and usage.

For aerospace or military application, life-saving, life-sustaining, safety critical applications or any application where failure may cause severe personal injury or death, please consult us before design-in the capacitor in your application.

Except for the written expressed warranties, CapXon does not impliedly, by assumption or whatever else, warrant, undertake, promise any other warranty or guaranty for any CapXon product.

For further information, please visit our website www.capxongroup.com or contact CapXon directly

All reel dimensions in mm

Ø D	4	5	5	6.3	6.3	6.3	6.3	8	8	8	8	10	10	10	10	12.5	16	16	18	18
L	5.5	5.5	5.8	5.5	5.8	6.1	7.7	6.5	7.7	10.5	11.7	8.7	10.5	12.4	16.5	14	17	21.5	16.5	21.5
A	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
B	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
C	14	14	14	18	18	18	18	18	18	26	26	26	26	26	26	34	46	46	46	46
D	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
E	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
F	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380

Remark: Standard = Paper reel

All reel dimensions in mm

Ø D	4	5	5	6.3	6.3	6.3	6.3	8	8	8	8	10	10	10	10	12.5	16	16	18	18
L	5.5	5.5	5.8	5.5	5.8	6.1	7.7	6.5	7.7	10.5	11.7	8.7	10.5	12.4	16.5	14	17	21.5	16.5	21.5
A	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2
B	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5	21.5
C	14	14	14	18	18	18	18	18	18	26	26	26	26	26	26	34	46	46	46	46
D	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
E	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
F	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380

Remark: Plastic reel = Add code "U" at the end of the part number

REVISION RECORDS

Ver.	Date	Process Number	Changed by	Description
001	2024-11-02	TBAKA-2411000002	Xu XiaoJun	Initial specification
002				
003				
004				
005				
006				

CUSTOMER APPROVAL

Date	Name	Department	Signature